

Jan Bauer

Homepage: japhba.github.io

EDUCATION

- Ph.D. student** in theoretical neuroscience 10/2022–
Gatsby Unit, University College London, United Kingdom
Edmond and Lily Safra Center, The Hebrew University of Jerusalem, Israel
Working on kernel descriptions of chaos in recurrent models of the brain.
Advisors: Agostina Palmigiano and Jonathan Kadmon
- Research stay** 11/2023–04/2024
Summerfield lab, University of Oxford, United Kingdom
Modelled cognitive flexibility with linear neural networks.
- M.Sc. in physics** GPA 1.0/1.0, with distinction 2019–2022
RWTH Aachen University, Germany
Focussed coursework on Statistical Mechanics of Neural Networks, Theoretical Neuroscience, Deep Learning in Physics Research, Quantum Information, and Computational Physics
- Master's thesis** 09/2021–08/2022
Juelich Research Center, Germany
Analysis of computation in cortical networks by Gaussian process regression.
Advisors: Prof. Moritz Helias, Dr. Christian Keup (now EPFL Switzerland)
- Bilateral graduate exchange program** 09/2019–03/2020
The University of Tokyo, Japan
Focussed coursework on Tensor Networks and Universal Biology, which investigates stem cell differentiation and evolution by a mathematical, complex systems approach. Member of Japanese university choir “The White Rose”.
- B.Sc. in physics** GPA 1.3/1.0, with distinction 10/2016–09/2019
RWTH Aachen University, Germany
- Bachelor's thesis** 03/2019–09/2019
Juelich Research Center, Germany
Description of unsupervised learning in Boltzmann machines via Feynman diagrams.
Advisors: Prof. Moritz Helias, Dr. Tobias Kuehn (now ENS Paris)
- 4x **teaching assistant** in theoretical physics and lab courses, mentoring 1st to 10th semester students.

PUBLICATIONS

- Bauer J.***, De Schamphelaere C.*, Karvonen A., Luick N., Nanda N., 2026. *Building Better Activation Oracles*. **ICML 2026 Mechanistic Interpretability Workshop** 
- De Schamphelaere C.*, **Bauer J.**, Nanda N., Ong E., 2026. *LoRAcles: Self-Supervised Weight-Space Interpretability at Scale*. **ICML 2026 Mechanistic Interpretability Workshop** 

Bauer J., Keup C., Kadmon J., Helias M., 2026. *Discrete signaling mediates chaotic regularization in recurrent neural networks*. **arXiv preprint** 

Bauer J., Fischer K., Helias M., Palmigiano A., 2026. *A unified theory of feature learning in RNNs and DNNs*. **ICML 2026** 

Schutzzeichel L., **Bauer J.**, Bouss P., Musall S., Dahmen D., Helias M., 2025. *Recurrent dynamics underlying transient neural representations*. **bioRxiv** 

Schwarcz J.*, **Bauer J.***, Rajabi, H., Marmur, G., Lottem, E., Kadmon, J., 2025. *Neural mechanisms of flexible perceptual inference*. **PLOS Comp. Biology** 

Rubruck J., **Bauer J.**, Saxe A., Summerfield C. *Neural mechanisms of flexible perceptual inference*. **CCN 2025 Conference proceedings**

Sandbrink K.* **Bauer J.***, Proca A.M.*, Saxe A., Summerfield C., Hummos A.* , 2024. *Flexible task abstractions emerge in linear networks with fast and bounded units*. **NeurIPS 2024 (Spotlight)** 

PROFESSIONAL EXPERIENCE

MATS 10.0 Fellowship

02/2026–08/2026

Berkeley, United States

Built meta-models ([AOs](#), [LoRAcles](#)) to audit LLMs when black-box monitors fail.

Mentor: Neel Nanda

PIBBSS Fellowship

06/2024–09/2024

London Initiative for AI Safety, London, United Kingdom

Mechanistic interpretability of neural geometry behind chain-of-thought, [public talk](#)

Mentor: Jan Hendrik Kirchner (Anthropic)

Research Internship

03/2021–06/2021

Bosch Center for Artificial Intelligence, Renningen, Germany

Co-developed and implemented architectural extensions of Bayesian version of the Neural Process model.

Advisor: Michael Volpp

HONOURS

HUJI International PhD Talent Scholarship

since 2023

Springorum memorial coin for outstanding graduates

2022

Scholarship fellow of the Konrad Adenauer Foundation (one of 13 academic talent promoters established by the Federal Ministry of Education and Research) 2016-2022

Fellow of RWTH Aachen University's **Dean's List** of top 5% students

since 2019

PRESENTATIONS

- Invited talk **Temporal feature learning**, Institute for Advanced Simulation, Juelich, Germany 10/2025
- Poster **Recurrent networks under constraint of sparse reward learn interacting belief state dynamics**, COSYNE 2024, Lisbon, Portugal 03/2024
- Talk **Can discrete neurons be useful?**, ELSC Annual Retreat, Nahsholim, Israel 06/2023
- Poster **Discrete communication mediates effective regularization in chaotic recurrent networks**, COSYNE 2023, Montréal, Canada 03/2023
- Random Matrix Theory for Machine Learning**, Parallel Sessions of INM-6 Annual Retreat 05/2022
- Inference with Graphical Models**, Book Club of Institute for Neuroscience and Medicine, Juelich Research Center 10/2021
- Proof-read and advised high school textbook** together with teacher Dr. Bardo Diehl at didactics congress MNU Aachen in front of 40 participants (*"Zentrale Experimente für das Abitur", Cornelsen 2017*) 04/2017

SCHOOLS & WORKSHOPS

- 12-day workshop "Analytical Connectionism"** 09/2023
University College London, United Kingdom
- 3-week summer school "Mathematical Methods in Computational Neuroscience"** 07/2023
NTNU, Norway
- 5-day workshop "Recent advances in understanding artificial and biological neural networks"** 02/2023
Les Houches School of Physics, France
- 5-day summer school "Reinforcement Learning"** 06/2022
Vrije Universiteit Amsterdam

TEACHING EXPERIENCE

- Statistical Physics and Field Theory summer 2022
- Theory of Electromagnetism winter 2022
- Preparatory math course for computer scientists 10/2020

Theory of Thermodynamics	winter 2020
Introductory lab course for physicists	09/2018
Introduction to Theoretical Physics (1 st year B.Sc.)	winter 2018

LEADERSHIP AND OUTREACH

Social commitment

Volunteer at the City of Aachen, supporting Egyptian family with homework and administrative tasks	2017–2020
Co-organisation of "LernFair"-AI lessons, part of a project aimed at high school students during the pandemic	winter 2021

Full-time scholarship by the Konrad Adenauer Foundation

Elected spokesperson of local group of 25 students	2019–2020
Increased participation of students by bundling proposals for engaging and meaningful group activities, such as a volunteer week in kindergarten in socially deprived suburb of Aachen	
Initiated and organized 4-day seminar on the scientific voice in democracies , with invited speakers on the philosophy of science, politics, recent societal challenges, and science communication	04/2022

[Service](#) to inform about cancellation of school lessons in the early morning before classes start 2015

Development of [mobile app](#) to create a precise elevation map of Aachen by use of barometer data of phones, targeted at finding a least elevation bicycle route 2016

[Application of machine learning and Fourier decomposition](#) to successfully remove chequered paper background from handwritten lecture notes 2020

SOFTWARE AND LANGUAGE PROFICIENCY

Software

Python, JAX, PyTorch

3D computer graphics with Blender

Languages

German (native)

English (academic proficiency, daily usage)

French (good, DELF A2)

Japanese (good, weekly practice with Tandem partner)

RECREATIONAL ACTIVITIES

Bicycle touring and medium distance running (olympic-distance triathlon in September 2022)

Singing in university choir (tenor voice)

REFERENCES

Dr. Agostina Palmigiano, Ph.D. advisor, Gatsby Unit, UCL

Dr. Jonathan Kadmon, The Hebrew University of Jerusalem

Prof. Moritz Helias, M.Sc. thesis advisor, Juelich Research Center

Dr. Michael Volpp, research internship advisor, Bosch Center for Artificial Intelligence

Dr. Christian Keup, M.Sc. thesis advisor, EPFL

Dr. Tobias Kuehn, B.Sc. thesis advisor, Ecole Normale Supérieure Paris

Updated: September 25, 2026